

swtpcsim — Quick Reference

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Getting out, and back in

Key	Does
<code>^E</code>	STOP — stop the machine and take the keyboard back. Nothing is lost.
<code>RUN</code>	Resume, at the exact instruction it stopped on.
<code>QUIT</code>	Leave. (There is no <code>EXIT</code> .)

Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values (`SET mem0 fill=` then `Tab`); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.swtpcsim_history`.

Key	Does
<code>Ctrl-A</code> / <code>Ctrl-E</code> (or <code>Home</code> / <code>End</code>)	start / end of line
<code>Alt-B</code> / <code>Alt-F</code> (or <code>Ctrl-Left</code> / <code>Ctrl-Right</code>)	back / forward one word
<code>Ctrl-W</code> / <code>Ctrl-K</code> / <code>Ctrl-U</code>	erase word behind / to end of line / whole line
<code>Backspace</code> / <code>Delete</code>	erase before / under the cursor

Command line

```
swtpcsim [options] [machine]
```

machine	a built-in name, or a config file (has a '/' or ends .toml). Omitted: ./swtpcsim.toml if there is one, else `default`.
-m, --machine <n>	ALWAYS a built-in name -- never a file.
-f, --file <path>	ALWAYS a file -- never a built-in name.
-n, --none	empty backplane: no boards, no memory, nothing.
-l, --list	list the built-in machines and exit.
-s, --script <f>	run a command script, then exit with its status. Paths in it are relative to the script's folder.
-x, --exec <cmd>	run one monitor command (repeatable), then exit.
-i, --interactive	after --script/--exec, stay in the monitor.
--mcp	MCP server on stdio.
-v, --version	print the version and exit.
-h, --help	print this help and exit.

Monitor commands

Type the part before the bracket.

Command	Does	Usage
BO[ARDS]	List, add, or remove boards on the backplane.	BOARDS [LIST] ADD <type> <id> [k=v...] REMOVE <id>
B[REAK]	Set a breakpoint on an address, memory/I/O access, or tape stop.	BREAK [<addr> MEM R W <addr> TAPE STOP] [IF <expr> LOADS <expr>] [TRACE ON OFF]
COM[PARE]	Compare a range of memory against another address.	COMPARE <range> <addr>
C[ONFIG]	Load or save the whole machine as a TOML file.	CONFIG LOAD <f.toml> CONFIG SAVE <f.toml>
CONN[ECT]	Attach a serial unit to an endpoint (console, socket, file, ...).	CONNECT <id>:<u> <endpoint>
CONS[OLE]	Show or set the host console's properties.	CONSOLE [<k>=<v>...]
DE[POSIT]	Write bytes into memory at an address.	DEPOSIT <addr> <bytes...>
DI[SASM]	Disassemble memory into instructions.	DISASM [<addr> <range>] [n] [CPU=6800 6809]
DISC[ONNECT]	Unplug the endpoint from a serial unit.	DISCONNECT <id>:<u>
D0	Run a file of monitor commands, one per line, as if typed.	D0 <file>
D[UMP]	Show memory as hex and ASCII.	DUMP [<addr> <range>] [WIDTH=16]
E[EDIT]	Enter bytes into memory interactively from an address.	EDIT <addr> [ROM]
EX[AMINE]	Point the front panel at an address (and show that byte).	EXAMINE [<addr>]
F[ILL]	Fill a range of memory with a byte.	FILL <range> <byte>
HE[LP]	Show help for a command.	HELP [<command>]
H[ISTORY]	Replay the recent instruction (or bus-cycle) history.	HISTORY [BUS CPU] [n]
L[OAD]	Load a file into memory (binary, Intel hex or S-record).	LOAD <file> [AT <addr>] [FORMAT=BIN HEX SREC] [ROM]
MA[CHINE]	Load a built-in machine by name (MACHINE none empties the backplane).	MACHINE <name> MACHINE none

M[OUNT]	Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it.	MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...]
MOV[E]	Copy a range of memory to another address.	MOVE <range> <dest> [ROM]
N[EXT]	Step one instruction, running any JSR/BSR to completion.	NEXT
NO[BREAK]	Remove a breakpoint, or all of them.	NOBREAK [id]
P[OWER]	Power-cycle the machine -- the only thing that clears RAM.	POWER
Q[UIT]	Leave the simulator.	QUIT
REGI[ON]	Add a RAM or ROM region to a memory board.	REGION ADD <id> type=ram rom at=<addr> [size= mount=]
RE[GS]	Show the CPU registers (SET REG changes one).	REGS SET REG <r>=<v>
RES[ET]	Reset the machine, keeping RAM (RESET CPU resets just the processor).	RESET [CPU]

REST[ORE]	Load machine state back from a snapshot.	RESTORE <file>
R[UN]	Start or resume the machine, optionally at an address.	RUN [addr]
SA[VE]	Write a range of memory out to a file.	SAVE <file> <range> [FORMAT=BIN HEX OCTAL PRN]
SEA[RCH]	Find bytes or a string in a range of memory.	SEARCH <range> <bytes...> "str"
SE[T]	Change a property of a board, the console, display, a register, the bus, or the machine.	SET <id>[:<u>] CONSOLE DISPLAY MACHINE REG BUS <k>=<v>
SH[OW]	Display the state of a board, the bus, or the machine.	SHOW <id> BOARDS BOARD <type> [UNITS] MACHINES MACHINE [<name>] BUS [MAP IRQ CONTENTION] ROMS MOUNTS PATHS CONSOLE DISPLAY SYMBOLS CLOCK VERSION
SN[APSHOT]	Save the whole machine state to a file.	SNAPSHOT <file>
STA[RTUP]	Edit the machine's boot list (the commands CONFIG SAVE writes as startup = [...]).	STARTUP [ADD <command> REMOVE <n> CLEAR]

S[TEP]	Run one instruction (or n), showing the registers after each.	STEP [n]
SY[MBOLS]	Load or clear a symbol table for disassembly.	SYMBOLS LOAD <file> [REPLACE] SYMBOLS CLEAR
T[RACE]	Log every bus cycle while the machine runs.	TRACE ON OFF [file] [MASK=IRQ,CONTENTION]
TY[PE]	Feed text to the guest as if typed at its keyboard.	TYPE "text"
U[NMOUNT]	Take a disk or tape out of a drive.	UNMOUNT <id>:<u>
W[H0]	Say which board answers an address or I/O port.	WHO <addr>

Boards

CPU

Type	What it is
6800	Altair 680b / SWTPC CPU board: a Motorola 6800
6809	CPU board: a Motorola 6809
mp09	SWTPC MP-09: a 6809 with the DAT and the S-BUG ROM

Memory

Type	What it is
memory	RAM/ROM board: plain, unbanked memory regions

Disk

Type	What it is
dc4	SWTPC DC-4: WD179x 5.25" floppy controller

Serial

Type	What it is
680io	Altair 680b onboard I/O: 6850 console and strap port
mps	SWTPC MP-S: one 6850 serial port on an SS-30 slot

Tape

Type	What it is
680kcacr	Altair 680b KCACR: Kansas City audio cassette

Parallel and printer

Type	What it is
680uio	Altair 680b Universal I/O: 6850 serial port and 6820 PIA

Machines

Machine	What it is
altair680	The Altair 680b -- MITS's second machine, and a different animal from the 8800.
swtpc	The SWTPC 6800 -- Southwest Technical Products' 1975 computer, the machine this simulator is named for.
swtpc09	The SWTPC 6809 -- an SS-50 system with the MP-09 processor board and S-BUG in its ROM.

A machine file, in one look

```
[machine]
name      = "mine"
base      = "default"      # start from a machine, and say what is DIFFERENT
startup   = ["RUN FF00"]    # the operator's own keystrokes. There is no BOOT verb.

[[board]]                # type + a NEW id      -> ADD the card
type = "mps"              # type + an id from the base -> REPLACE it outright
id   = "mps0"              # NO type + an id      -> MODIFY the one already there
at   = 8004                 # remove = true       -> PULL THE CARD OUT

[board.unit.tty]          # a unit's own settings
connect = "console"

[[board.region]]          # a list the card owns (memory)
type = "ram"
at   = 0000                 # hex: it is an address
size = "32K"               # decimal: it is a size

[[board.drive]]           # a list the card owns (disk controllers)
unit  = 0
mount = "flex.dsk"         # relative to THIS FILE

[console]                 # the HOST's terminal -- not a board
strip7out = true
base      = octal           # read/print the wire class in split octal
```

Paths: a path *inside* a machine file is relative to **that file**. A path you *type* is relative to **your shell**.

Endpoints — `CONNECT <id>:<unit> <endpoint>`

Endpoint	Is
<code>console</code>	the host terminal. Exactly one unit may hold it.
<code>null</code>	nowhere. Writes vanish, reads never come.
<code>loopback</code>	itself — what you write comes back.
<code>scripted</code>	a caller in place of a human — what MCP and the tests type into.
<code>socket:PORT</code>	listens , as a raw pipe — for a program at the far end. <code>?banner</code> greets each caller.
<code>socket:HOST:PORT</code>	calls out , as a raw pipe.
<code>telnet:PORT</code>	listens , speaking Telnet — this is telnet-in for a person. Greets each caller; <code>?banner=off</code> stops it.
<code>telnet:HOST:PORT</code>	calls out , taking the telnet client's part.
<code>serial:DEVICE</code>	a real serial port on this host.
<code>in:PATH</code>	a host file as a reader (paper tape). <code>?cps=N</code> paces it.
<code>out:PATH</code>	a host file as a punch — 8-bit clean, never truncating.
<code>terminal</code>	a window the simulator draws itself (SDL builds). <code>?emulation=vt100 adm3a vt52 h19</code> , <code>?size=COLSxROWS</code> .
<code>printer:QUEUE</code>	a real print queue on this host.
<code><endpoint> FILE</code>	a tap: append <code> FILE</code> to any endpoint to also log the line.
<code><endpoint> socket:PORT</code>	a live mirror: <code>telnet</code> in to watch and take over. <code>?ro</code> = watch-only.